Operating Systems (OS-MIRI)
Facultat d'Informàtica de Barcelona
OS-MIRI exam, January 12th 2021

Answer the questions concisely and precisely in the same sheet
The multiple-choice test is closed-book
For the exercises you are allowed a single legible double-sided cheat sheet
Duration: 1 hour, 45 minutes

Name and Surname:

---

1. (5 pts) Select the answer (only one) that you consider to be correct on every section. Every correct answer computes 1/2 points. Every wrong answer deducts 1/6.

   (a) Which of the following scheduling policies provides better load balancing?
   - ◯ Time-sharing with local queues and balanced mapping of new processes to queues
   - ◯ Time-sharing with local queues
   - √ Time-sharing with a single global queue
   - ◯ Time-sharing with a single global queue and predefined hard affinity

   (b) A service is implemented using *double modular redundancy* (with 2 replicas). If the mean time until each replica crashes is 1 week and a failure typically takes 7 hours to fix, what is the availability of the replicated service?
   - √ 99.84%
   - ◯ 99.826%
   - ◯ 96%
   - ◯ 95.833%

   Solution: Probability of one replica not available = MTTR / (MTTF + MTTR) = 7/(7*24 + 7) = 0.04 ; overall availability = $1 - 0.04^2 = 0.9984$

   (c) Which of the following statements about *hardware-assisted virtualization* is false?
   - ◯ The hardware adds root (for the hypervisor) and non-root (for the guests) modes of operation and enables transitions between them, namely VM entries (from hypervisor to guest) and VM exits (from guest to hypervisor)
   - ◯ The hardware provides state areas to save/load the guest and host processor states on VM exits/entries
   - ◯ The hardware implements control structures to configure the behavior of some instructions when executed by the guest
   - √ The hardware scans the guest for basic blocks and rewrittes them to replace sensitive instructions with code that emulates their effect against the guest state

   (d) Which of the following statements about *pre-copy VM migration* is false?
   - ◯ The migration may never end if the VM dirties pages faster than they can be transferred
   - ◯ A given page can be transferred multiple times if it is dirtied repeatedly at the source host during the migration
   - √ Once the VM starts running on the destination host, the pages which have not been yet received are faulted in from the source host over the network
   - ◯ If the destination host fails during the migration, the VM can be recovered from the source host

(e) Which of the following statements about I/O virtualization without hardware support is true?

- ◯ Device drivers in the guest can program DMA transfers without the intervention of the hypervisor
- √ Device drivers in the guest can program DMA transfers, but they must be intercepted by the hypervisor, who will program them after remapping the guest physical addresses provided by the drivers into host physical addresses
- ◯ Device drivers in the guest can program DMA transfers, but they must be intercepted by the hypervisor, who will program them by using the physical memory addresses provided by the drivers
- ◯ Device drivers in the guest must be rewritten to remove DMA transfers, because only the hypervisor can program them

(f) In a distributed datacenter with 8 nodes where data is kept consistent thanks to a quorum protocol, which of the following quorums (where R is number of readers and W is number of writers) is free of write-write conflicts.

- √ R=2 W=5
- ◯ R=3 W=4
- ◯ R=4 W=3
- ◯ R=7 W=2

(g) Imagine the scoreboard of a baseball game (like Terry13), where the current score is 0-0 (home=0 and visitors=0). This scoreboard is maintained in a data store with two replica nodes using an *eventual consistency with a read your writes* guarantee. Two different users, Alice and Bob, execute the following sequence of events:

	Alice	**Bob**
(1)	W(home = 1)	W(visitor = 1)
(2)	W(home = 2)	W(visitor = 2)
(3)		W(visitor = 3)
(4)	R(home)	
(5)	R(visitor)	

Which of the following results (H-V) may not be read by Alice from this scoreboard?

- √ 1-0
- ◯ 2-0
- ◯ 2-1
- ◯ All previous values are valid, a not valid one is .....

(h) In a datacenter that uses a distributed POSIX filesystem using a stateful server. Which of the following operations are done at server side when a client issues a *read*() operation?

- ◯ open, lseek, read, close
- ◯ open, read, close
- ◯ lseek, read
- √ read

(i) How many processes related to the Hadoop DFS are running in a single node installation?

- ◯ 5
- √ 3
- ◯ 2
- ◯ 1

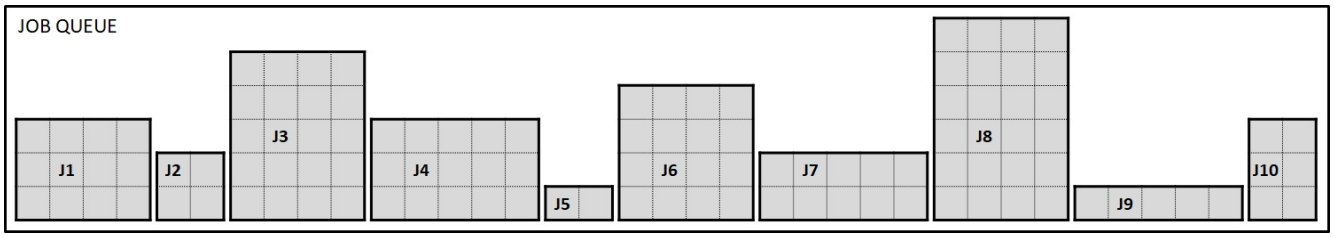
Solution: 3 : datanode, namenode, and secondary-namenode

(j) Which of the following management approaches is not used to prevent 'hot spots' in a WSC?

- ◯ Spreading the tasks to balance the load of the servers
- ◯ Delaying task start or extending task execution time to smooth the load
- ◯ Placing high-load tasks in the servers of the WSC that are easier to cool
- √ Using energy storage to deal with short demand peaks or short-term supply reductions

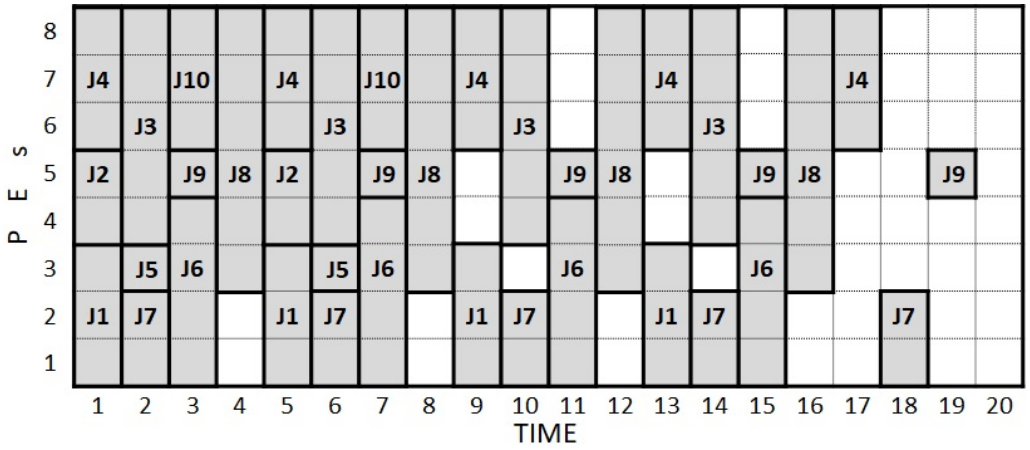
Name and Surname:

2. (1.25 pts) Some jobs are currently waiting in the queue to be scheduled in a WSC with 8 PEs. Each job shows graphically the number of requested PEs (vertical axis) and its duration (horizontal axis).

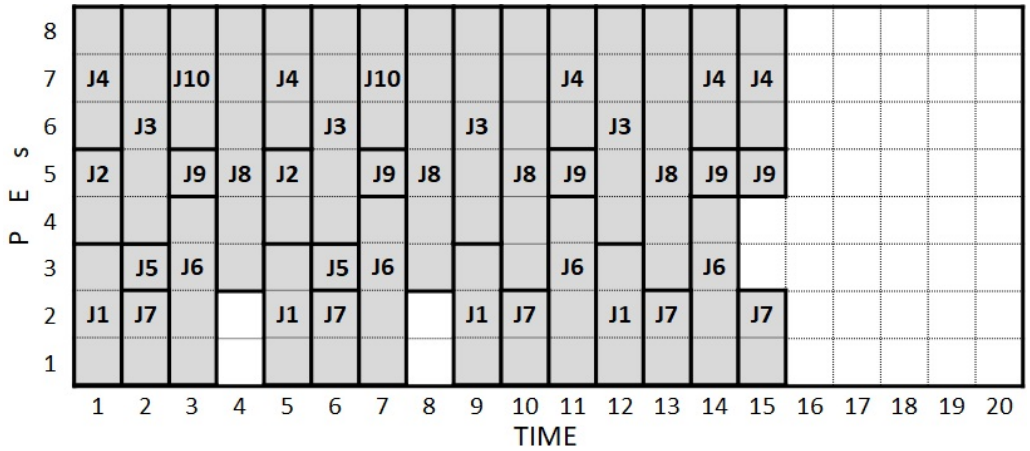


a) Indicate how the jobs are scheduled by using *Gang Scheduling*, which is configured with a quantum of one time unit, uses 'Left-Right First Fit' to allocate jobs to slots (jobs in the queue are scanned sequentially; free slots are scanned sequentially and the first one with sufficient capacity to run the job is chosen; if no free slot has sufficient capacity, a new slot is used; odd slots are filled starting from PE=1; even slots are filled starting from PE=8), and implements *Affinity Scheduling* (the PEs allocated to a job are the same during its entire execution), if:

i) There is not any re-packing of the jobs upon every job termination.



ii) There is a re-packing of all the jobs upon every job termination. Slots are unified as much as possible, but maintaining jobs affinities.



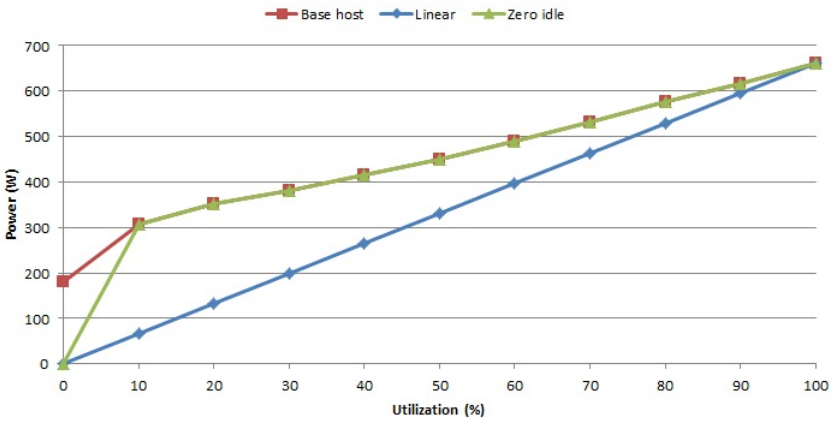
b) Calculate the *slowdown* for the last job to finish in previous scenario i), and its corresponding *slowdown* in scenario ii).

Solution: Slowdown J9 scenario i) $= (T_W + T_R)/T_R = (14 + 5)/5 = 3.8$
Slowdown J9 scenario ii) $= (T_W + T_R)/T_R = (10 + 5)/5 = 3$

3. (1.25 pts) Consider a host with the performance (in millions of operations per second, Mop/s) and power (in watts, W) profiles at different utilization levels shown in the next table:

utilization (%)	0	10	20	30	40	50	60	70	80	90	100
performance (Mop/s)	0	291	581	869	1160	1449	1741	2031	2320	2611	2889
power (W)	181	308	351	382	416	451	490	533	576	617	662

a) Justify whether this host is energy-proportional (hint: compare it graphically with a linear system with idle power = 0 W and peak power = 662 W). What if the idle power (at 0% utilization) of the host was zero?



Solution: Energy-proportional systems should consume no power when idle and gradually consume more power as the activity level increases. As shown in the plot, this host consumes significant power when idle, and the power consumption is higher than in a proportional system, especially for low utilizations. Eliminating the idle power helps when the system is not utilized, but there is still higher power consumption when utilization > 0%

b) Consider a WSC made of 1000 hosts like the one described above, which runs a given workload for 12 minutes. For simplicity, assume a discrete utilization distribution across the 1000 hosts, with 108 hosts running at 0% utilization, 80 hosts at 10% utilization, etc. as shown in column '# hosts' below. Calculate the total work executed (in billions of operations, Gop), the total energy consumed (in watts-hour, Wh), and the Data Center energy Productivity (DCeP). Compare i) a baseline configuration that keeps the unused hosts idle, and ii) a low-power configuration that suspends the unused hosts instead of keeping them idle, knowing that:

- utilization while suspending a host: $U_{on \to off} = 20\%$
- utilization while resuming a host: $U_{off \to on} = 60\%$
- time needed to suspend a host: $\delta_{on \to off} = 20$ s
- time needed to resume a host: $\delta_{off \to on} = 90$ s
- power consumption while the host is suspended: $P_{off} = 9$ W

Solution: time to run workload: $T = 12$ minutes $= 720$ s $= 0.2$ h
power consumption while suspending a host: $P_{on \to off} = 351$ W
power consumption while resuming a host: $P_{off \to on} = 490$ W
energy to suspend a host: $E_{on \to off} = P_{on \to off} \cdot \delta_{on \to off} = 351 \cdot 20 = 7020$ J $= 1.95$ Wh
energy to resume a host: $E_{off \to on} = P_{off \to on} \cdot \delta_{off \to on} = 490 \cdot 90 = 44100$ J $= 12.25$ Wh
time while the host is suspended: $T_{off} = T - \delta_{on \to off} - \delta_{off \to on} = 720 - 20 - 90 = 610$ s
energy while the host is suspended: $E_{off} = P_{off} \cdot T_{off} = 9 \cdot 610 = 5490$ J $= 1.525$ Wh

Name and Surname:

total work (Gop) = perf (Mop/s) · T (s) · #hosts / 1000
energy i) (Wh) ($U \geq 0$) = energy ii) (Wh) ($U > 0$) = power (W) · T (h) · #hosts
energy ii) (Wh) (U = 0) = ($E_{off} + E_{on \to off} + E_{off \to on}$) · #hosts
DCeP = $\sum_{U=0}^{100}$ total work (Gop) / $\sum_{U=0}^{100}$ energy (Wh)

U (%)	perf (Mop/s)	power (W)	# hosts	total work (Gop)	energy i) (Wh)	energy ii) (Wh)
0	0	181	108	0	3909.6	1698.3
10	291	308	80	16761.6	4928	
20	581	351	153	64002.96	10740.6	
30	869	382	246	153917.28	18794.4	
40	1160	416	191	159523.2	15891.2	
50	1449	451	115	119977.2	10373	
60	1741	490	51	63929.52	4998	
70	2031	533	21	30708.72	2238.6	
80	2320	576	15	25056	1728	
90	2611	617	12	22559.04	1480.8	
100	2889	662	8	16640.64	1059.2	
		Total	1000	673076.16	76141.4	73930.1
				DCeP	8.84	9.10

c) Assume now that the middleware of the WSC is able to consolidate the workload, which results in the utilization distribution across the 1000 hosts shown in column '# hosts' below. Calculate again the total work executed (in billions of operations, Gop), the total energy consumed (in watts-hour, Wh), and the Data Center energy Productivity (DCeP) considering configurations i) and ii) described before, and justify which of the four configurations evaluated (b.i, b.ii, c.i, c.ii) is more energy-efficient.

U (%)	perf (Mop/s)	power (W)	# hosts	total work (Gop)	energy i) (Wh)	energy ii) (Wh)
0	0	181	504	0	18244.8	7925.4
10	291	308	6	1257.12	369.6	
20	581	351	8	3346.56	561.6	
30	869	382	11	6882.48	840.4	
40	1160	416	26	21715.2	2163.2	
50	1449	451	57	59466.96	5141.4	
60	1741	490	95	119084.4	9310	
70	2031	533	123	179865.36	13111.8	
80	2320	576	76	126950.4	8755.2	
90	2611	617	40	75196.8	4936	
100	2889	662	54	112324.32	7149.6	
		Total	1000	706089.6	70583.6	60264.2
				DCeP	10.00	11.72

Solution: Consolidating the workload and suspending unused hosts (configuration c.ii) is the most energy-efficient configuration because it has the highest DCeP = 11.72.

(this page left blank intentionally)

Name and Surname:

4. (1.25 pts) Regarding a Redis system:

   a) What is the name of the general protocol used to keep the data consistent across the system?

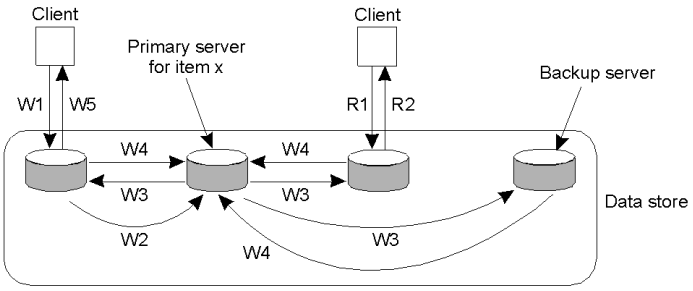
   b) Explain this protocol briefly and sketch a small diagram of the components and the sequence of events involved.

   c) Relate this protocol to any configuration/option you have used in the laboratory.

Solution:

a) Redis uses a **primary based remote-write protocol**.

b) The following figure sketches this protocol with the different replicas and a couple of clients:



W1. Write request
W2. Forward request to primary
W3. Tell backups to update
W4. Acknowledge update
W5. Acknowledge write completed

R1. Read request
R2. Response to read

All write operations are centralized in a primary replica (always the same). The primary replica is in charge of coordinating write operations to the data item. In Redis, write operations on a different replica than the primary are disabled (therefore all of them are local and the forwarding to the primary is never needed).

Write Requests from clients at primary replicas are acknowledged and propagated in the background to the remaining replicas.

c) In the lab, you may configure when a request is acknowledged by changing a special parameter 'min-replicas-to-write' that specifies how many replicas must be written before the write succeeds.

Read requests may go to any replica which will answer with the current value, meaning the most recent one or an older one if the write operation has not been updated yet.

5. (1.25 pts) We have a Cassandra cluster compound by 2 datacenters of 4 nodes each. The replication factor at each datacenter is configured to 3. In this scenario:

   a) Sketch the architecture of the cluster and name and describe briefly the main components involved.

   b) Explain the general behavior to keep the data consistent when a client issues a write request relating it to the components stated before.

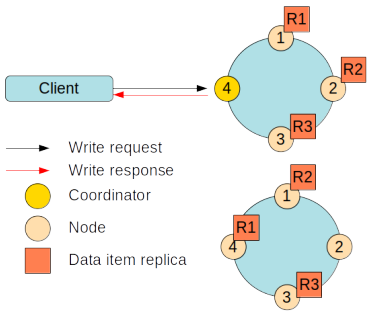
   c) For the case where a consistency level of ONE is used, explain this consistency level, the number of network messages involved in the previous write request and how many are required to acknowledge the request.

   d) The same but for the consistency level of QUORUM.

   e) Imagine that two nodes of a single datacenter have crashed, what would happen to the write request in the previous case? And if instead of the crashed nodes, the switch connecting both datacenters crashes?

Solution:

a) The following figure represents the cluster with 2 datacenters and 4 nodes each:



The figure also shows:

(a) the **CLIENT**, which will issue a write request.

(b) the **COORDINATOR**, who will manage the consistency level of the data item.

(c) and the **REPLICAS**, that keep a copy of data items

b) When a write request is received in a node, it becomes the **COORDINATOR**, and manages the consistency level of the data item. It contacts the **REPLICAS** of this particular data item asking for the write request and after receiving a number of succesful answers according to the current consistency level, acknowledges the write request.

c) A consistency ONE implies that a single replica must succeed to acknowledge the request. All data item replicas are contacted with the write request, therefore 6 (3+3) messages are sent, and 6 answers from the replicas are sent back, from which the first arriving one will acknowledge the request. In summary, 12 messages are involved, and 7(6+1) are required.

d) A consistency of QUORUM means that a quorum of replicas must succeed to acknowledge the request. This quorum must be half of the sum of the replication factors plus one. In this case $4 = (3+3)/2 + 1$. This means that all replicas will be contacted (6), and 6 answers will come back. In summay, 12 messages are involved, and 10 (6+4) are required.

e) In the crashed nodes case, there will be enough safe nodes alive (8), and therefore the request will succeed, but in the other case we have a network partition , only 4 nodes are available and therefore the request will not succeed.